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Non-destructive analysis of chlorine in fly ash cement concrete

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ABSTRACT

Preventive measures against reinforcement corrosion in concrete require increasing concrete density to prevent the diffusion of chloride ions to the steel surface. Pozzolanic materials, such as fly ash (FA), silica fume (SF), and blast furnace slag (BFS) are added to concrete to increase its density. Monitoring the chloride concentration in concrete is required to assess the chances of reinforcement corrosion. In this study, FA was added to Portland cement concrete to increase its density. Prompt gamma neutron activation analysis (PGNAA) technique was utilized to analyze the concentration of chlorine in concrete. The chlorine concentration in the FA cement concrete was evaluated by determining the yield of 1.16, 1.95, 6.11, 6.62, 7.41, 7.79, and 8.58 MeV gamma-rays of chlorine from the FA concrete specimen containing 0.4–3.5 wt% chlorine. An excellent agreement was noted between the experimental yield of the prompt gamma-rays and the calculated yield obtained through the Monte Carlo simulations. The Minimum Detectable Concentration (MDC) of chlorine in FA cement concrete was also calculated. The best value of MDC limit of chlorine in the FA cement concrete was found to be 0.022 ± 0.007 and 0.038 ± 0.017 wt% for 1.16 and 6.11 MeV prompt gamma-rays, respectively. Within the statistical uncertainty, the lower bound of MDC meets the maximum permissible limit of 0.03 wt% of chlorine in concrete set by American Concrete Institute Committee 318.

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1. Introduction

Chloride-induced reinforcement corrosion is the principal cause of deterioration of concrete structures worldwide [1–4]. There are several ways by which chlorides can be introduced into concrete. They may exist at small levels within the mix ingredients of concrete (e.g., aggregates), or they may be introduced by the use of a chloride-containing admixture (e.g., calcium chloride that is normally used as an accelerator) or use of saline water as mixing water (when fresh water is not available). However, chlorides in concrete most often come from the service environment to which the concrete structures are exposed (e.g., exposure to deicing salt or a marine environment).

One of the preventive measures against reinforcement corrosion is to increase concrete density. Pozzolanic materials, such as fly ash (FA), silica fume (SF), blast furnace slag (BFS), etc. are added to concrete, as a partial replacement of Portland cement, to make it dense and hence mitigate reinforcement corrosion [3,4]. The pozzolanic material reacts with calcium hydroxide that is liberated during the hydration of cement to form supplementary cementing material that makes concrete further dense. Hence, it is desirable to control the chemical composition of the pozzolanic material in concrete such that the required elements are available for the pozzolanic reaction. Further, monitoring the concentration of chloride ions in a blended cement concrete is highly desirable to minimize the chances of reinforcement corrosion. In this study, fly ash was added to Portland cement to make the concrete dense. As shown in Table 1, FA contains silica (52.3 wt%), alumina (25.2 wt%), iron oxide (4.60 wt%), lime (10.0 wt%), MgO (2.20 wt%), and SO_3 (0.60 wt%). Fly ash is a fine, glass-like powder recovered from gases in a coal-fired electric-power generation facility. It is a pozzolanic substance containing aluminous and siliceous material that forms cement in the presence of moisture. When mixed with lime and water, it forms a compound similar to Portland cement [3]. Prompt Gamma Neutron Activation Analysis (PGNAA) technique has proven to be an effective tool to determine the chloride concentration in concrete non-destructively [5–10]. An accelerator-based PGNAA monitor for assessing the elemental composition of concrete, both plain and blended, is under development at King Fahd University of Petroleum and Minerals (KFUPM) [7–9]. Previously, the chloride concentration was measured in the plain cement concrete utilizing the PGNAA setup at KFUPM [8,9]. In this paper, the results of a study conducted to evaluate the chlorine concentration in FA cement concrete are reported. The reported study is part of a comprehensive project currently undertaken by the authors to

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Table 1
Chemical composition (wt%) of Portland cement and fly ash.

Constituent	Type I cement (wt%)	Fly ash (wt%)
SiO_2	20.52	52.30
Al_2O_3	5.64	25.20
Fe_2O_3	3.80	4.6
CaO	64.35	10.0
$CaCO_3$	–	–
MgO	2.11	2.20
SO_3	2.1	0.60
K_2O	0.36	0.10
Na_2O	0.19	0.10

develop response data of a D (d,n) accelerator-based PGNAA setup for various types of concretes utilized by the construction industry. These data would be utilized to design a portable PGNNA technique-based monitor for assessing concrete composition in the field and monitoring reinforcement corrosion.

2. Prompt gamma measurements of chlorine concentration in FA cement concrete

The chlorine concentration in the FA cement concrete specimens containing 0.4–3.5 wt% chloride was measured using the PGNAA setup [7–9]. The concrete specimens were prepared by mixing 20 wt% FA as a replacement of cement. All the concrete ingredients were thoroughly mixed and thereafter poured in a specially designed 14-cm-long and 12.5-cm-radius molds. Seven FA cement concrete specimens 12.5 cm in radius and 14 cm in length were prepared with 0.4, 0.8, 1.5, 2.0, 2.5, 3.0, and 3.5 wt% chloride following the procedure described elsewhere [7–9]. The concrete specimens were demolded after 1 day, and, thereafter they were cured by covering them with wet burlap for 13 days and finally dried in an electric oven at 70 °C for 2 days.

The chloride-contaminated FA cement concrete specimens were irradiated at the PGNAA setup of the KFUPM 350 kV electrostatic accelerator [7–9]. A pulsed beam of 2.8 MeV neutrons was produced via D(d,n) reaction. Deuterium target used in the present study consisted of deuterium adsorbed in titanium layer deposited on a thin copper backing. Due to continuous implantation of deuterons by incident beam in target backing, target thickness was not well defined but neutron intensity at the target was estimated to be $3–5 \times 10^5$/s. The deuteron pulse had a width of 5 ns and a frequency of 31.25 kHz. The typical beam current of the accelerator was 1–3 μA.

The prompt gamma-rays produced in the concrete specimens were detected by a 25 cm × 25 cm (diameter × height) NaI detector. The prompt gamma-ray data were acquired for a preset charge measured at the electrically isolated neutron-producing copper target enabling current measurement. The charge was measured by a calibrated Beam Current Integrator (BCI). The prompt gamma-ray detector signals were acquired in coincidence with a gate signal being derived from the beam pick-up. The chlorine prompt gamma-ray data were acquired with 4-cm-thick front moderator of the PGNAA setup.

3. Results and discussion

Fig. 1 shows prompt gamma-ray spectra from FA cement concrete specimens containing 0.8, 1.5, 2.5, and 3.5 wt% chlorine above 3 MeV gamma-ray energy. The full energy (F) and single escape (S) peaks of the prompt gamma-rays are marked in Fig. 1. The large peak on the right end of the spectrum is the gamma-ray peak from activation of sodium ^{23}Na and iodine ^{127}I in a NaI detector [10]. As a result of neutron capture in these elements, ^{24}Na and ^{128}I are produced that subsequently decay and emit their characteristic gamma-rays. Iodine has a characteristic gamma-ray energy of 6.828 MeV while Na has a characteristic energy of 6.959 MeV. These gamma-rays are observed only in the activation of large-sized NaI detectors since a small-sized detector has lower detection efficiency for these gamma-rays [10]. Gardner et al. [10] have observed full-energy peak at 6.885 MeV energy along with a single escape peak at 6.4 MeV due to ^{24}Na and ^{128}I in the activation of 5 in × 5 in (diameter × height) and 6 in × 6 in detectors. This peak energy was attributed to the weighted average peak of two unresolved peaks of ^{24}Na (6.828 MeV) and ^{128}I (6.959 MeV) [10], as was done in the present work, since a cylindrical 10 in × 10 in (height × diameter) NaI detector was used.

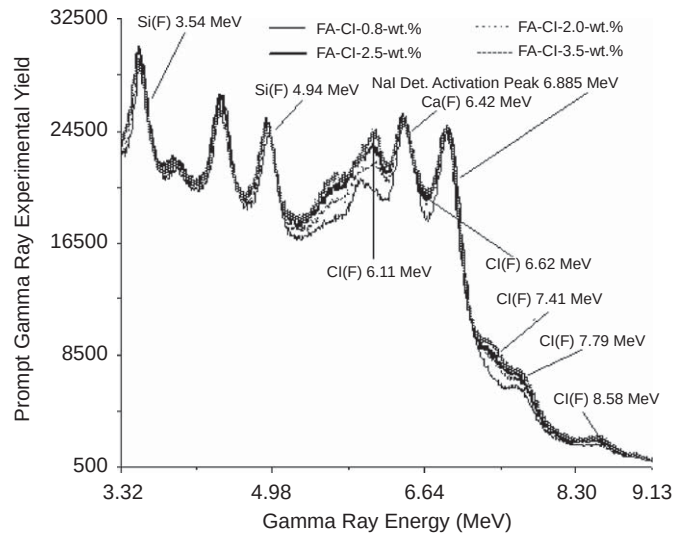

Fig. 1. Prompt gamma-ray spectra from four chloride-contaminated FA cement concrete specimens containing 0.8, 2.0, 2.5, and 3.5 wt% chlorine with energies in excess of 3 MeV.

Table 2
Energies and partial elemental cross-section $\sigma_\gamma^z(E_\gamma)$ barns of capture gamma rays in concrete [11].

Element	Gamma-ray energy (MeV)	$\sigma_\gamma^z(E_\gamma)$-barn
Calcium	1.942	0.352
	4.418	0.0708
	6.420	0.176
Silicon	3.539	0.1190
	4.934	0.1120
Aluminum	1.779	0.232
	7.724	0.0493
Iron	7.631	0.653
	7.646	0.549
Hydrogen	2.223	0.3326

Additionally, Fig. 1 also shows prompt gamma-rays from chlorine superimposed on prompt gamma-rays from FA cement concrete. The partial elemental cross-section in barns $\sigma_\gamma^z(E_\gamma)$ for the production of gamma-rays E_γ from various elements Z in concrete assuming natural abundance is given in Table 2 [11] while the prompt gamma-ray partial elemental cross-sections in barns $\sigma_\gamma^z(E_\gamma)$ for chlorine are listed in Table 3 [11]. The peak on the right-hand side of the NaI detector activation peak (unlabeled) is an unresolved doublet of Fe peaks at 7.631 and 7.646 MeV. The calcium and silicon prompt gamma-ray peaks are located on the left-hand side of the NaI detector peaks. The full-energy peak of

Table 3
Energies and partial elemental cross-section $\sigma_{\gamma}^{z}(E_{\gamma})$ barns of capture gamma rays of chlorine [11].

Gamma-ray energy (MeV)	$\sigma_{\gamma}^{z}(E_{\gamma})$-barn
1.164	8.91
1.951	6.33
1.959	4.10
2.863	1.82
3.061	1.13
4.979	1.23
5.715	1.82
6.110	6.59
6.619	2.53
6.627	1.47
7.413	3.29
7.790	2.66
8.578	0.883

calcium Ca(F) at 6.42 MeV is overlapping with the single escape peak of the detector activation peaks. The double escape peak from calcium (not labeled) is located immediately after the single escape peak Ca(S) of calcium corresponding to full-energy peak at 6.42 MeV. Fig. 1 also shows full-energy prompt gamma-ray peaks from silicon Si(F) at 4.94 and 3.54 MeV and a peak at 4.44 MeV, which includes the single escape events from 4.94 MeV photons. This peak was also present in the background spectrum taken without the concrete specimen. Moreover, its intensity is higher than that of the Si(F) peak at 4.94 MeV , thus suggesting an additional contribution from a different photon source. This contribution was attributed to 4.44 MeV gamma-rays from an unshielded Am–Be neutron source that was located nearby.

Several prompt gamma-rays are emitted by chlorine due to capture of thermal neutrons. In this study, only chlorine prompt gamma-rays at 1.16, 1.95, 6.11, 6.62, 7.41, 7.79, and 8.58 MeV were analyzed. The partial elemental cross section in barns $\sigma_{\gamma}^{z}(E_{\gamma})$ for the production of gamma rays E_{γ} for chlorine are listed in Table 3 [11]. The main feature of data in Fig. 1 is the increased intensities of some peaks due to the interference of chlorine gamma-rays with those of concrete. The 7.413 MeV gamma-rays from chlorine have strong interference with the detector activation peak, and single escape peak of 7.413 MeV gamma-rays overlaps with the 6.885 MeV peak from activation of NaI detector. The 6.619 MeV gamma-ray peak from chlorine (8% intensity) is located between the NaI activation peak at 6.885 MeV peak, and the full-energy peak of 6.42 MeV gamma-ray from calcium. Also, the full-energy peak of the 6.619 MeV chlorine peak partially overlaps with the full-energy peak of the 6.420 MeV calcium peak. The full-energy peak of 6.11 MeV gamma-ray interferes with the escape peak of 6.42 MeV gamma-ray from calcium.

The chlorine gamma-ray intensity from each of the chloride-contaminated FA cement concrete was obtained after subtracting the normalized prompt gamma-ray spectra from pure FA cement concrete specimen. The data were normalized to the same amount of integrated beam charge measured at the target. Fig. 2 shows the subtracted spectra of chlorine prompt gamma-ray from FA cement concrete specimens containing 0.8, 1.5, 2.5, and 3.5 wt% chlorine. Fig. 3 shows the prompt gamma-ray spectra from FA cement concrete specimens containing 0.8, 2.0, 2.5, 3.0, and 3.5 wt% chlorine below 2.4 MeV. The main features of Fig. 3 are the 2.22 MeV hydrogen capture peak and two full-energy peaks at 1.16 and 1.95 MeV due to capture of thermal neutrons in chlorine. Fig. 4 shows the background subtracted spectra of 1.16 and 1.95 MeV chlorine prompt gamma-rays from FA cement concrete containing 0.8, 2.0, and 3.0 wt% chlorine. The gamma-rays corresponding to 1.16 and 1.95 MeV are clearly shown in Fig. 4.

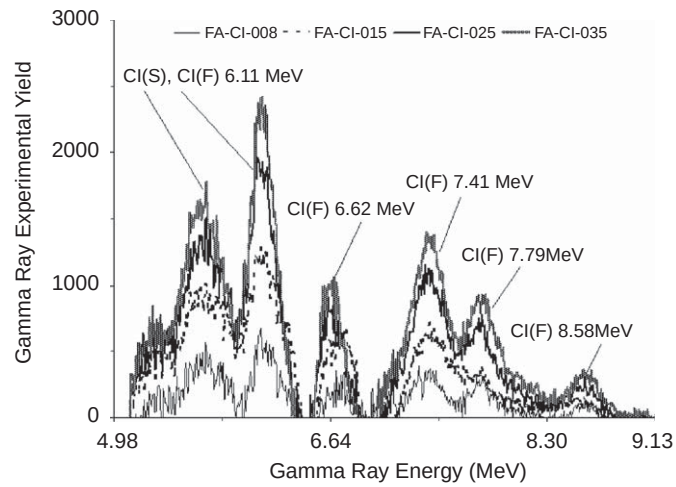

Fig. 2. Difference spectra of chlorine with energies in excess of 3 MeV gamma energy from FA cement concrete specimens containing 0.8, 1.5, 2.5, and 3.5 wt% chlorine.

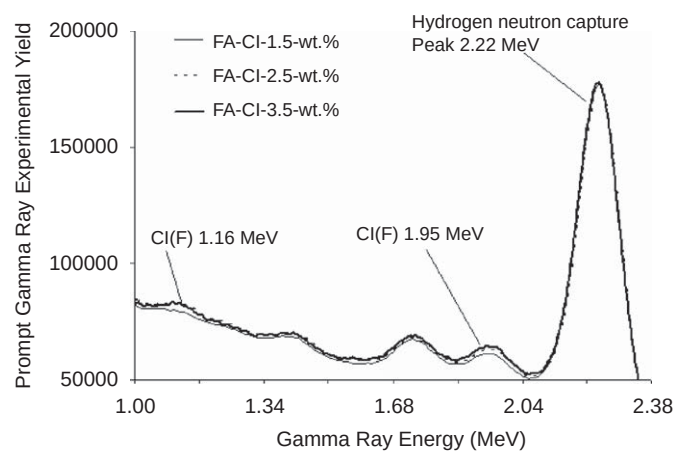

Fig. 3. Prompt gamma-ray spectra from chloride-contaminated FA cement concrete specimens containing 1.5, 2.5, and 3.5 wt% chlorine with energies below 2.4 MeV gamma energy.

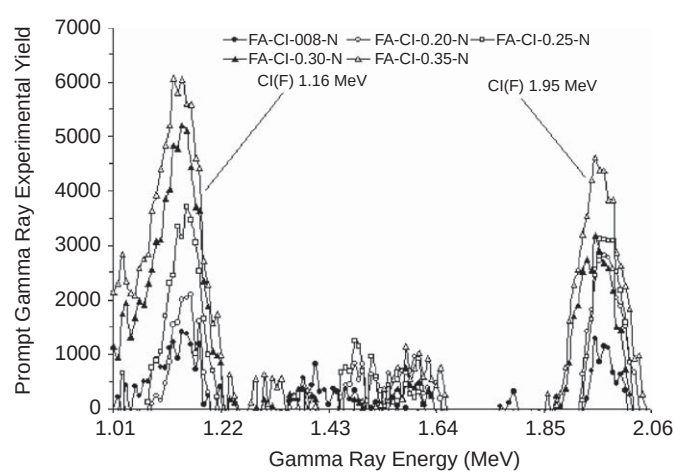

Fig. 4. Difference spectra from FA cement concrete specimens containing 0.8, 2.0, 2.5, 3.0, and 3.5 wt% chlorine energies below 2.4 MeV gamma energy.

The counts under each peak were integrated from all seven different-spectra of chlorine containing FA cement concrete specimens. Fig. 5 shows the normalized experimental yield of 1.16 and 1.95 MeV gamma-rays as a function of chlorine

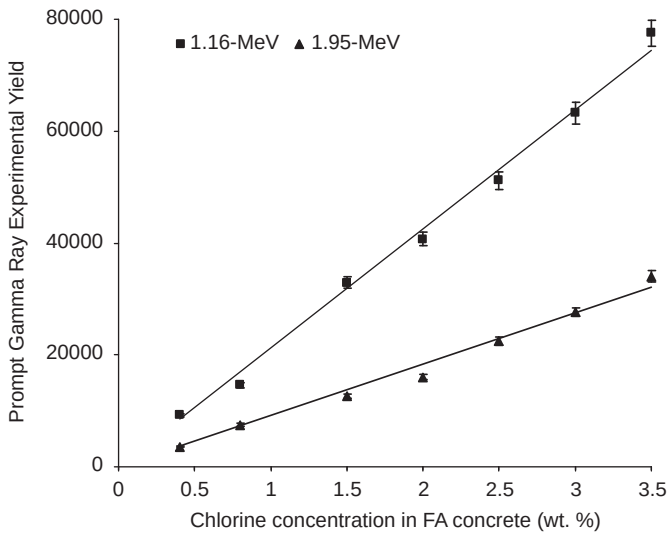

Fig. 5. Experimental yield of 1.16 and 1.95 MeV prompt gamma-rays from chlorine in the FA cement concrete specimens containing 0.4–3.5 wt% chlorine plotted as a function of chlorine concentration. The solid line shows normalized calculated yield of the gamma-rays obtained through Monte Carlo calculations.

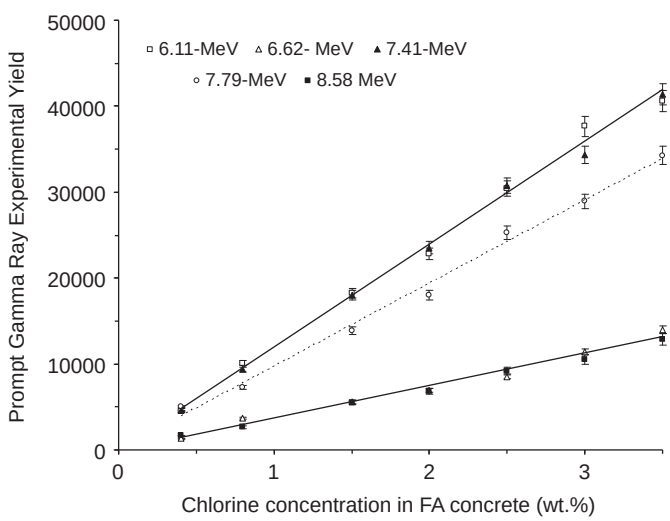

Fig. 6. Experimental yield of 6.11, 6.62, 7.41, 7.79, and 8.58 MeV prompt gamma rays from chlorine in the FA cement concrete specimens containing 0.4–3.5 wt% chlorine plotted as a function of chlorine concentration. The solid line shows normalized calculated yield of the gamma rays obtained through Monte Carlo calculations.

concentration in the FA cement concrete while data on yield of 6.11, 6.62, 7.41, 7.79, and 8.58 MeV prompt gamma-rays from chlorine in the seven chloride-contaminated FA cement concrete specimens are plotted in Fig. 6. The uncertainties shown in the data are mainly statistical in nature. Within the experimental uncertainties, there is a linear correlation between the 1.16, 1.95, 6.11, 6.62, 7.41, 7.79, and 8.58 MeV prompt gamma-ray yield and chlorine concentration. The solid lines in Figs. 5 and 6 show the normalized calculated yield of the prompt gamma-rays from chlorine in FA cement concrete obtained through Monte Carlo simulations.

The minimum detectable concentration (MDC) of chlorine in FA cement concrete was calculated using the procedure described in Ref. [8]. The detection limit for an elemental concentration MDC, measured under a peak with net counts N_P and associated background counts N_B (under the peak), is defined by the relation [7]:

$$\mathrm{MDC} = 4.653(C/N_P)\sqrt{N_B}$$

where C is the element's concentration in the peak. The error in MDC is $\sigma_{\mathrm{MDC}} = (C/N_P)[\sqrt{(2N_B)}]$.

Table 4
Minimum detectable concentration (MDC) of the KFUPM PGNAA setup for chlorine in plain and fly ash cement concrete specimens.

E_γ (MeV)	MDC (wt%)	
	Plain cement concrete [7]	Fly ash cement concrete
1.16	0.077 ± 0.025	0.022 ± 0.007
1.94	–	0.051 ± 0.015
5.72	0.255 ± 0.050	–
6.11	0.140 ± 0.068	0.038 ± 0.017
6.62	–	0.059 ± 0.027
7.41	0.075 ± 0.023	0.050 ± 0.027
7.79	–	0.076 ± 0.029
8.58	–	0.068 ± 0.029

Table 4 shows the MDC of chlorine in FA cement concrete specimens determined for the KFUPM PGNAA setup with MDC values for pure cement concrete measured in a previous study [7]. The MDC of chlorine in FA cement concrete specimens for the KFUPM PGNAA setup varies over the range 0.022–0.076 wt% while for the pure concrete it varied from 0.075 ± 0.023 to 0.255 ± 0.050 [7]. The improvement in MDC value for FA cement concrete is due to improvement in data analysis and background subtraction technique used in the present measurements. The maximum permissible limit of chloride concentration in concrete according to the American Concrete Institute Committee 318 is 0.03 wt% [2]. The best values of MDC limits of chlorine in the FA cement concrete were found to be 0.022 ± 0.007 and 0.038 ± 0.017 wt% for 1.16 and 6.11 MeV prompt gamma-rays, respectively. Within the statistical uncertainty, the lower bound of MDC meets the maximum permissible limit of 0.03 wt% of chlorine set by ACI standards [2].

Based on the data shown in Figs. 5 and 6, it can be concluded that the KFUPM accelerator-based PGNAA setup [7–9] can be used successfully for non-destructive analysis of chlorine in FA cement concrete.

4. Conclusions

The concentration of chlorine in fly ash cement concrete was determined using an accelerator-based PGNAA setup. The experimental yield of 1.16, 1.95, 6.11, 6.62, 7.41, 7.79, and 8.58 MeV gamma-rays from chlorine in FA cement concrete specimens was measured as a function of chlorine concentration. An excellent agreement was noted between the gamma-ray experimental yield and the calculated yield obtained from the Monte Carlo simulations. The best value of MDC limit of chlorine in the FA cement concrete was found to be 0.022 ± 0.007 and 0.038 ± 0.017 wt% for 1.16 and 6.11 MeV prompt gamma-rays, respectively. Within the statistical uncertainty, the lower bound of MDC meets the maximum permissible limit of 0.03 wt% of chlorine set by ACI Committee 318 [2].

The results of this study have shown that the KFUPM accelerator-based PGNAA setup could be utilized to assess the concentration of chlorine in the FA cement concrete specimens.

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